



Certified LambdaProof Administrator (CAA) Training Curriculum

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Administered By: LambdaProof Certification Working Group

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Program Overview

The Certified LambdaProof Administrator (CAA) designation validates that operational staff possess the knowledge and skills required to manage LambdaProof-certified enterprise deployments. This certification is mandatory for organizations seeking Tier 3 (LambdaProof-Certified Enterprise) status, which requires a minimum of 2 Certified LambdaProof Administrators on staff.^[1]

Certification Levels

Level	Designation	Target Role	Duration	Prerequisites
Foundation	CAA-F	Operations Staff, Support Engineers	2 days	None

Professional	C/A-P	System Administrators, DevOps	4 days	C/A-F
Expert	C/A-E	Security Architects, Compliance Leads	3 days	C/A-P + 6 months experience
Instructor	C/A-I	Internal Trainers	2 days	C/A-E + teaching assessment

Module 1: Constitutional Foundations (C/A-F)

1.1 Λ Proof Architecture Overview

- The Λ -Constitution and its immutable provisions^[1]
- Meta-Theorem of Prime Identity (MTPI) principles^[2]
- Prime-epoch gating and drift-bound enforcement ($\delta < 0.3$)^[2]
- Zero-surveillance by design philosophy^[3]

1.2 Conscious Sovereignty Layer (CSL) Operators

- **Neutrality Operator (N):** Uniform treatment verification—same input produces same output regardless of subject identity^[1]
- **Beneficence Operator (B):** Consent requirement enforcement and data minimization^[1]
- **Silence Clause (S):** NO-OP when neutrality or beneficence fails; no action taken, audit event logged^[1]
- **Commutation Invariant:** $N \circ B \circ I = B \circ N \circ I$ (order independence)^[1]

1.3 Core Protocol Components

- Λ RootContract interface and state transitions^[1]
- Λ -Archivum audit event schema and hash chaining^[3]
- Capability Token structure, signature verification, and expiry enforcement^[1]
- Intent Schema canonicalization and policy binding^[1]

Assessment: 50-question multiple choice exam (70% pass threshold)

Module 2: Operational Administration (C/A-F → C/A-P)

2.1 Key Management

- HSM or equivalent for signing keys^[1]
- Key rotation procedures (minimum annual)^[1]
- Multi-party key ceremony for root keys^[1]
- Key recovery procedures and compromise response planning^[1]

2.2 Access Control Implementation

- Role-based access control (RBAC) configuration^[1]
- Privileged access management^[1]
- Access logging and monitoring setup^[1]
- Quarterly access reviews procedures^[1]

2.3 Deployment Security

- Infrastructure-as-code principles^[1]
- Immutable deployments (no production changes)^[1]
- Signed releases with verification^[1]
- Vulnerability scanning in CI/CD pipelines^[1]

2.4 Conformance Test Suite Operation

- Running the test suite: `npm install @lproof/conformance-tests`^[1]
- Test categories: CORE, CSL, GATE, PROTO, ADAPT, INTEROP^[1]
- Generating certification reports^[1]
- Interpreting test results and remediation^[1]

Assessment: Hands-on lab (4 hours) + written exam (60 questions, 75% pass threshold)

Module 3: Monitoring & Incident Response (CAA-P)

3.1 Real-Time Monitoring

- CSL violation alerting configuration^[1]
- Audit log integrity monitoring^[1]
- Anomaly detection for abuse patterns^[1]
- Performance and availability monitoring^[1]

3.2 Audit Log Management

- Event schema: XiStateTransition, PrimeGateActivation events^[1]
- Hash chain verification: `EventN.prev_event_hash = Event(N-1).event_hash`^[3]
- Merkle checkpoint construction and blockchain anchoring^[3]
- Privacy guarantees (commitments only, no PII)^[1]

3.3 Incident Response Procedures

- Defined severity levels and SLAs^[1]
- Communication templates and escalation procedures^[1]
- Post-incident review process^[1]
- Annual tabletop exercises^[1]

Assessment: Tabletop incident simulation + written exam

Module 4: Compliance & Certification (CAA-P → CAA-E)

4.1 Certification Tier Requirements

Tier	Requirements	Validation
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Tier 1 (Λ Proof-Compatible)	100% conformance tests, no patent claims on core	Automated + self-attestation ^[1]
Tier 2 (Λ Proof-Verified)	Tier 1 + security audit + CSL compliance review	Third-party audit + WG review ^[1]
Tier 3 (Λ Proof-Certified Enterprise)	Tier 2 + on-site audit + staff certification	On-site audit + continuous monitoring ^[1]

4.2 Domain-Specific Certifications

- **Λ Proof-Health:** HIPAA mapping, consent PLIC validation, PHI handling audit^[1]
- **Λ Proof-Finance:** KYC/AML workflow verification, transaction audit trail^[1]
- **Λ Proof-Agent:** Capability token enforcement, tool adapter security^[1]
- **Λ Proof-Device:** Low-power implementation review, firmware attestation^[1]

4.3 Audit Preparation

- Documentation requirements^[1]
- Security audit scope (CSL enforcement, cryptography, prime gating, token security)^[1]
- Working with approved auditors (Trail of Bits, NCC Group, Least Authority, etc.)^[1]

Assessment: Mock audit scenario + compliance documentation exercise

Module 5: Advanced Topics (C/A-E)

5.1 Zero-Knowledge Proof Operations

- Miller-Rabin primality verification in circuits^[2]
- Drift-gate state transitions^[2]
- Evidence validation pipelines (ZK proofs, attestations, receipts)^[1]
- Post-quantum readiness considerations^[1]

5.2 Adapter Coordination

- Adapter Coordination Interface (ACI) implementation^[1]
- Fail-closed enforcement patterns^[1]
- Token binding to canonical intent hash^[1]
- Replay protection mechanisms^[1]

5.3 Governance Participation

- MIP (MTPI Improvement Proposal) process^[1]
- Constitutional constraints and Ethics Guardian review^[1]
- Working Group participation^[1]
- Appeals process procedures^[1]

Assessment: Capstone project + oral examination

Examination & Certification

Exam Structure

Level	Written Exam	Practical Lab	Project
C/A-A-F	50 questions (70% pass)	—	—
C/A-A-P	60 questions (75% pass)	4-hour hands-on	—
C/A-A-E	40 questions (80% pass)	6-hour simulation	Capstone
C/A-A-I	Teaching demonstration	—	Curriculum development

Recertification Requirements

- **Annual:** Complete 16 hours of continuing education^[1]
 - **Biennial:** Pass recertification exam (condensed format)
 - **Specification Updates:** Complete delta training within 90 days of major releases
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Fee Schedule

Certification	Exam Fee	Training (Optional)
CAA-Foundation	\$500	\$1,500 (2 days)
CAA-Professional	\$750	\$3,000 (4 days)
CAA-Expert	\$1,000	\$2,500 (3 days)
CAA-Instructor	\$500	\$1,500 (2 days)

Fee Waivers: Available for open-source projects, academic institutions, and non-profits per the standard waiver policy.^[1]

Training Delivery Options

- **Self-Paced Online:** Video modules, interactive labs, practice exams
 - **Live Virtual:** Instructor-led sessions via video conference
 - **On-Site Corporate:** Custom training at enterprise locations (minimum 8 participants)
 - **Bootcamp Intensive:** 5-day immersive program (CAA-F through CAA-P)
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Contact Information

- **Training Registration:** training@lproof.org
 - **Certification Inquiries:** certification@lproof.org
 - **Corporate Programs:** enterprise@lproof.org
 - **Curriculum Feedback:** curriculum@lproof.org
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This curriculum ensures that Certified AProof Administrators possess comprehensive knowledge of both the constitutional foundations and operational requirements for managing AProof-certified enterprise deployments.^{[3][1]}

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1. [lets-produce-a-list-of-documen-MocV13l6RI.gUHQs0MYElw.md](#)
 2. LProof-IP-Patent-Research.pdf
 3. lets-discuss-licensing-open-so-HV_bbj5uQlOPuXCF7b61YQ.md
 4. ΛProof Token Issuance.pdf
 5. Adapter Coordination Interface Specification.pdf
 6. Audit Logs_ Event Schema & Sanitized Examples.pdf
 7. ΛProof Intent Schema Specification.pdf
 8. Conscious Sovereignty Layer (CSL) Solver.pdf
 9. ΛProof Protocols.pdf
 10. DAO Charter.pdf
 11. Building a ΛDapp_ A Developer's Implementation Guide.pdf
 12. Capability Token Semantic Mapping.pdf
 13. Data Sovereignty Informational Personhood.pdf
 14. ΛProof_ An Architectural Blueprint for Verifiable Artificial Intelligence.pdf
 15. Λ^P-Archivum.pdf
 16. Patent-Search-Results.pdf
 17. LProof-IP-Strategic-Blueprint.pdf
 18. Governance for a Non-Surveillance Digital Ecosystem.pdf
 19. Token Specification.pdf
 20. Risk Classification.pdf
 21. Policy Rules Specification.pdf
 22. Evidence Validation.pdf
 23. How Trust is Measured.pdf